



ENGG4800 - Practical Four

Task Information

Robin Lane and top management have approved the schedule generated at the end of Part 3. As the project schedule is now approved, it can be set as a baseline – from which point the schedule can be progressed through and deviated from.

Useful Terms

Here is a reminder of useful terminology.

Term	About
PV / BCWS	Planned Value, also known as Budgeted Cost of Work Scheduled (BCWS)
EV / BCWP	Earned Value, also known as the Budgeted Cost of Work Performed (BCWP)
AC / ACWP	Actual Cost, also known as Actual Cost of Work Performed (ACWP)
SV	Schedule Variance in Cost Terms (BCWP – BCWS) it is the difference in cost between the current progress and the baseline activity.
CV	Cost Variance is the difference between how much the activity should have cost and how much it actually cost (BCWP - ACWP).
EAC	Estimate at Completion, expected total cost of activity.
BAC	Baseline Cost, the total planned cost for an activity.
CPI	Cost performance index. Typically, $BCWP \div ACWP$.
PCI	Percent Complete Indices

Tasks

Write a small memo responding to questions.

PART ONE – Setting the Baseline

1. Set the project (outcome of practical 3) as a Baseline.
 - *'Project → Set Baseline → For: Entire Project → Okay'*
2. What is the whole project estimated to cost?
 - *'Project → Project Information → Statistics'*
3. Which **activity** is estimated to cost the most to complete?
 - *'View → Gantt'* and then *'View → Data → Tables → Cost'*
4. Which **resource** commands the greatest total cost?
 - *'View → Resource Sheet'* and then *'View → Data → Tables → Cost'*
5. Use this resource: <https://www.mpug.com/create-monthly-cash-flow-report-microsoft-project-2013/> to create a cash flow report. Looking at the net costs of each activity, does this align with question 3?
6. What costs are not captured in this budget?

PART TWO – Many months later.

Robin Lane has requested a written status report for the Zuma project. For this task you will assume that today's date is August 16, 2064. The table given summarises the information regarding activities accomplished to date. **We will start the status report today but finish it in the next practical session.**

Table 1: Actual (differing from baseline) dates and durations of activities. (mm/dd/yy)

Activity	Start Date	Finish Date	Actual Duration	Remaining Duration
Market Analysis	1/2/64	2/1/64	23	
Product Design	2/4/64	3/20/64	34	
Manufacturing Study	3/21/64	4/22/64	23	
Product Design Selection	4/23/64	5/13/64	15	
Manufacturing Process	8/1/64	-	11	25
Detailed Product Design	5/14/64	7/31/64		
Test Prototype	8/1/64	8/15/64		

1. Set the current date to one day before our current day (16th August 2064).
 - 'Project → Project Information → Current Date: 1 5/8/64' (if your project is in dd/mm/yy)
2. Update the actual starts, finish dates, durations as per table 1.
 - 'Task → Gantt → View → Tables → Tracking'
 - Enter the information. (use the date format that your project is set to use)
3. Generate a table containing the following for your status report: PV, EV, AC, BAC, SV, CV, and CPI.
 - 'View → Other Views → Task Sheet' and then 'Tables → More Tables → Earned Value'
 - Other columns can be added.
4. How is the project progressing in terms of cost and schedule? Which tasks have gone well?
 - 'View → Tables → Variance'
5. What do the projects overall PCI_B and PCI_C indicate?
 - $PCI_B = EV/BAC$
 - $PCI_C = AC/EAC$
 - These will need to be hand calculated (for only the overall project as specified)

In the next practical we will continue this status report by:

- Determining the forecast costs at completion, EAC_f
- Determining the predicted VAC_f
- Interpreting the TCPI.
- Reporting the estimated completion date.
- Updating Robin Lan on the project in terms of its priorities (matrix)
- Provide an image of the Gantt chart (with the tracking table).

Outcomes

- Memo detailing the answers to the questions given under *Tasks*